



AVAILING **MOBILE TESTING**

ACROSS MULTIPLE PLATFORMS, DEVICES AND NETWORKS

Rendering superior customer experiences with cutting-edge
QA testing of mobile applications

Q qaoncloud.com



Mobile Application Testing

Testing ensures that mobile apps run optimally regardless of the OS or device used. The performance of mobile apps impacts user experience, which is an essential driver of brand value.

End-to-end testing of mobile applications contributes to the delivery of a robust and customer-centric experience for new users. To put it simply, the better the user experience, the higher the probability of increased outreach.

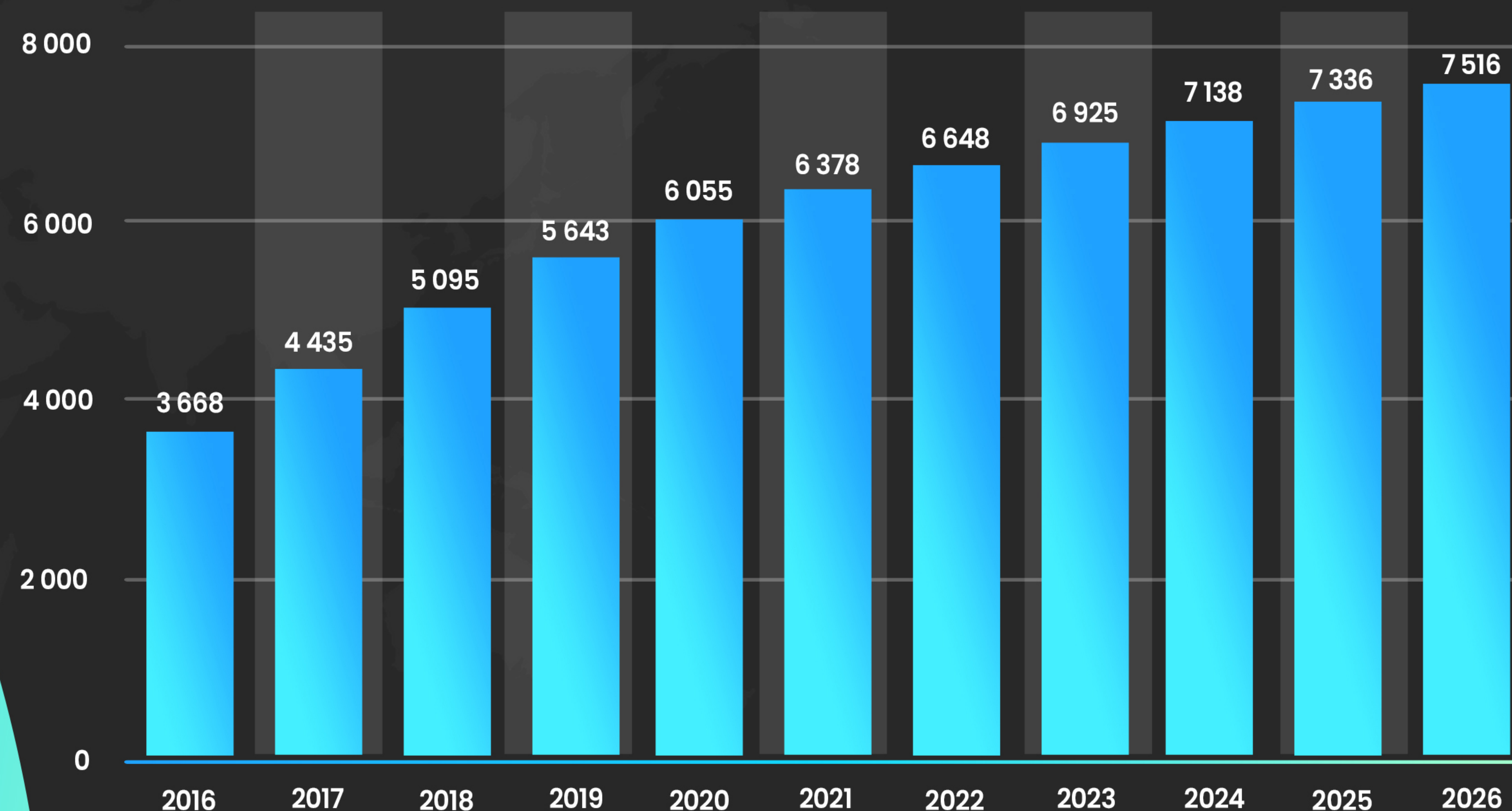
Mobile app testing is critical to

- Ensure seamless functionality and user experience across platforms and devices
- Optimize the performance of apps across different operating systems and versions
- Ensure consistency of functionality across multiple devices
- Make sure that the app's functionality, GUI, and performance are robust



Growing Mobile Market

- The number of mobile users globally is expected to hit **7.5 billion** by 2026
- There are **9000+** distinct mobile devices, and end-users in the real world can install apps on any of these devices



Such astronomical growth will inevitably lead to the development of advanced mobile apps to make business services more accessible.

The global mobile application market is expected to grow in revenue to **\$613** billion by **2025**, almost twice the market's worth in **2020**. **60%** of that revenue is from the mobile games industry.

However, businesses will also have to ensure that their mobile apps are optimised and future-ready for advanced smartphones. Stringent mobile testing and optimisation on the latest devices are critical for businesses to succeed in this endeavour.

Growing Customer Expectations

In proportion to the mobile market's growth, customer demands are also skyrocketing. Customers expect great user experiences, seamless performance and secure connectivity.

-  **80%** of customers delete or uninstall an app that doesn't meet their expectations
-  **50%** of customers tend to uninstall an app that occupies too much space on the device
-  **48%** of customers uninstall or abandon an app if it is slow

With **54.8%** of global website traffic coming from mobile devices and **89.2%** of device time spent within mobile apps

Criteria to Consider while Testing Mobile Applications

How mobile apps differ from web applications

Mobile apps differ from web applications in the way they are designed. The former is specifically designed and developed for operating on mobile devices, while the latter comprises software applications that reside on web servers and are accessed through web browsers.

Mobile apps need to be exclusively tested as they are :

- Designed for mobile devices and so have to be tested across numerous mobile devices
- Have different user interactions based on the touch screen on mobile devices like scrolling, zoom, pinching
- Downloaded on multiple devices and Operating systems



Challenges in Mobile Testing

According to the CapeGemini Quality Report (Mobile Testing), **18%** of responding companies say that they do not have enough time to test mobile apps, and **65%** do not have the right tools.

Compatibility across devices

Mobile Application Testing is also challenging due to compatibility issues. A mobile application can be deployed across devices which have different

- Operating systems like iOS, Android, etc
- Versions of an operating system such as Android Oreo, Android 10, 11, etc



Different types of mobile apps

Mobile apps could be of various types like Native, Hybrid, etc. Each of these has its unique testing strategies as the app behaviour from installation to functionality differs from one type to another.

Decision on automation tool for Mobile testing

The choice of tool poses its own limitations as the complexity of mobile test automation can be daunting as it depends on the nature of the app, device deployment details, etc. The decision on how much to automate requires experience and a balance between cost and effort.

Strategy for effective mobile app testing

- Clarity on the extent of device fragmentation, appropriate tools and frameworks.
- Effective planning on cross browser testing, unit testing, etc
- Simulated and real user condition testing to avoid unsatisfactory user experiences
- Decision on whether Beta Testing is needed
- Involvement of QA teams from early phases of development

Emerging Trends in Mobile Apps

5G

- Due to advantages like increased performance speed, fast data transfer, and lower latency, mobile apps are fast moving towards supporting 5G connections

IoT and Cloud

- IoT and the cloud offer advantages like reduced operational costs, improved efficiency, and increased connections to other platforms via APIs

AI and Machine Learning

- Mobile apps are turning to AI and Machine learning to aid navigation, speech recognition, and natural language processing (NLP)
- Behavioural algorithms based on AI also enhance security by analysing user behaviours and detecting fraud, suspicious activities, and potential information breaches

AI and Machine Learning

- With online shopping becoming more common than ever, more retailers are building native apps or progressive web apps (PWAs) to provide a better shopping experience

PWAs and Instant Apps

- PWAs are becoming more popular as they have benefits like reduced dependency on internet connections, faster loading times, automatic updates, and faster development time

AR/VR

- As AR/VR technologies advance, they will drive a transformation of gaming, entertainment, travel, and live streaming mobile apps. These technologies offer ways to simulate buying experiences at a virtual level and are the retail solutions of the future

With the landscape of mobile apps changing drastically, it is crucial to ensure testing stays relevant to provide seamless, secure user experiences.



Why Invest in a Mobile App Testing Partner

Compatibility across devices

- Experienced testing professionals can help reduce cost, resources, and time through optimized testing methods
- Mobile testing service providers can help access the latest tools and technologies without breaking the bank to invest

Compatibility across devices

- Efficient mapping of specific test schedules based on testing type, test scenarios, and the scope of automation
- Better quality of testing and quicker time is taken to test the system, due to core knowledge of having worked on a similar product
- Knowledge and application of best testing practices, frameworks, and test automation tools leading to a reduction in the overall testing time
- Efficient testing across domains, especially for niche requirements, with full test coverage using the latest testing tool

Improved Business Reputation

- Testing by experienced professional testing teams can ensure testing is carried out in an accurate and unbiased manner
- Robust testing leads to fewer bugs and has a direct positive impact on brand image and reputation

Better Understanding of the Latest Trends and Technologies

- Mobile testing service providers provide access to the industry's proven tools, and technologies and keep coming up with ideas for continuous improvement in software testing for better outcomes by leveraging their vast experience in the testing domain
- Mobile testing by a competent and qualified testing partner can result in cost-effective, structured, and efficient testing. A well-tested product has fewer performance issues and creates elevated customer experiences which translate into brand value for organization

Why QAonCloud

At QAoncloud, we perform end-to-end mobile testing efficiently, cost-effectively and meticulously to render error-free mobile apps with intended functionality and no privacy issues.



Methods of Mobile Testing

Application Testing

We test the mobile version of any website or software developed for different hand-held devices like smartphones, tablets, PCs or phones, etc. It can be games, animation videos, or any type of application

Device Testing

We ensure the quality of the mobile device in Device Testing by testing both hardware and software of the mobile

Our Comprehensive Mobile Testing Services

We use mobile simulators and emulators to test your applications. These are software programs that provide simulation for essential features of smartphones etc. Emulators are very useful at the early stage of the development process. An emulator replicates the device hardware and operating system and allows the tester to test the application on a desktop computer. Simulators are less complex applications that simulate the internal behaviour of a device.

Our mobile testing services comprise of testing that occurs in simulated environments post which our teams also perform extensive testing on real devices in actual user environments.



Functional Testing



Usability Testing



Compatibility Testing



Interface Testing



Security Testing



Installation Tests



Get Your App Tested By a
Remote, Flexible, On-Demand Team



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